Deutsche Akkreditierungsstelle

# Deutsche Akkreditierungsstelle

# Annex to the Accreditation Certificate D-PL-14316-01-01 according to DIN EN ISO/IEC 17025:2018

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| **Valid from:** | **13.01.2026** | **Valid to:** | **12.01.2031** |
| Date of issue: | 13.01.2026 | | |

**This annex is part of the Accreditation Certificate D-PL-14316-01-00.**

Holder of the Accreditation Certificate:

**Royal International Inspection Laboratory (RIIL)**
**Labs building, Sokhna port, Ain El Sokhna**
**SUEZ 43511, EGYPT**

The testing laboratory meets the requirements of DIN EN ISO/IEC 17025:2018 to carry out the conformity assessment activities listed in this annex. The testing laboratory meets additional legal and normative requirements, if applicable, including those in relevant sectoral schemes, provided that these are explicitly confirmed below.

The management system requirements of DIN EN ISO/IEC 17025 are written in the language relevant to the operations of testing laboratories and they conform to the principles of DIN EN ISO 9001.

Tests in the fields:

**physical, physico-chemical and chemical testing of food, feed and food contact materials microbiological testing of food, feed and environmental samples, equipment and utensils in the sector of food; molecular testing of food and feed**

**Within the indicated test areas the testing laboratory is permitted without being required to prior inform and obtain approval from DAkkS**

**[Flex A]** **to use standards or equivalent testing methods listed here with different issue dates.**

**[Flex B]** **to have the free choice from standardised or equivalent test methods.**

**The test methods listed are examples. The testing laboratory has an up-to-date list of all test methods within the flexible scope of accreditation. The list is publicly available on the website of the testing laboratory.**

**Content**

## 1 Physical, physico-chemical and chemical analysis of food, feed and food contact materials

### 1.1 Sample preparation [Flex A]

| | |
|---|---|
| ISO 12966-2<br>2017-03 | Animal and vegetable fats and oils - Gas chromatography of fatty acid methyl esters - Part 2: Preparation of methyl esters of fatty acids |
| EN 645<br>1993-10 | Paper and board intended to come into contact with foodstuffs - Preparation of a cold water extract |
| EN 647<br>1993-10 | Paper and board intended to come into contact with foodstuffs - Preparation of a hot water extract |

### 1.2 Determination of ingredients in food, feed and food contact materials by gravimetric analysis [Flex B]

| | |
|---|---|
| AOAC 920.100<br>2023-01 | Ash in tea<br>(Modification: *here also applicable for other food and feeding stuff*) |
| AOAC 920.102<br>2023-01 | Fiber (Crude) in tea |
| AOAC 920.153<br>2023-01 | Ash of meat<br>(Modification: *here also applicable for other food and feeding stuff*) |
| AOAC 920.169<br>2023-01 | Fiber (crude) in spices |
| AOAC 923.03<br>2023-01 | Ash of flour - Direct method<br>(Modification: *here also applicable for other food and feeding stuffs*) |
| AOAC 941.12<br>2023-01 | Ash of Spices - Gravimetric Methods<br>(Modification: *here also applicable for other food and feeding stuffs*) |
| AOAC 950.46<br>2023-01 | Loss on Drying (Moisture) in meat<br>(Modification: *here also applicable for other food and feeding stuffs*) |
| AOAC 962.09<br>2023-01 | Fiber (crude) in animal feed and pet food. Ceramic Fiber Filter Method |

| | |
|---|---|
| AOAC 975.12<br>2023-01 | Ash (acid-insoluble) of cacao products<br>(Modification: *here also applicable for other food and feeding stuffs*) |
| AOAC 991.36<br>2023-01 | Fat (crude) in meat and meat products |
| ISO 18609<br>2000-08 | Animal and vegetable fats and oils. Determination of unsaponifiable matter. Method using hexane extraction |
| EN 1186-2<br>2022-07 | Materials and articles in contact with foodstuffs - Plastics - Part 2: Test methods for overall migration in vegetable oils |
| EN 1186-3<br>2022-08 | Materials and articles in contact with foodstuffs - Plastics - Part 3: Test methods for overall migration in evaporable simulants |

### 1.3 Determination of ingredients in food and feed by titrimetric analysis [Flex B]

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|---|---|
| AOAC 920.160<br>2023-01 | Saponification Number (Koettstorfer Number) of Oils and Fats: Titrimetric Method. |
| AOAC 991.20<br>2023-01 | Nitrogen (Total) in milk - Kjeldahl methods<br>(Modification: *here also applicable for other food and feeding stuffs*) |
| AOAC 947.05<br>2023-01 | Acidity of Milk: Titrimetric Method |
| AOAC 965.33<br>2023-01 | Peroxide Value of Oils and Fats: Titration Method. |
| ISO 660<br>2020-03 | Animal and vegetable fats and oils — Determination of acid value and acidity |
| ISO 3657<br>2023-07 | Animal and vegetable fats and oils — Determination of saponification value |
| ISO 3961<br>2024-12 | Animal and vegetable fats and oils — Determination of iodine value |
| ES 63-10<br>2006-05 | Methods of analysis and testing for meat and meat products - Part 10: Determination of Thiobarbituric acid (TBA) |
| ES 2760-1<br>2017-05 | Physical and chemical methods for testing fish and fishery Products - Part 1: Frozen fish. 5.2 Determination of total volatile basic nitrogen |

### 1.4 Determination of ingredients in food and food contact materials by photometric analysis [Flex B]

| | |
|---|---|
| CEN/TS 13130-23<br>2005-02 | Material and articles in contacts with foodstuffs - Plastics substances subject to limitation - Part 23: Determination of formaldehyde<br>(Restriction: *no determination of HMTA*) |
| ES 2760-1<br>2017-(AMD 2018) | Physical and chemical methods for testing fish and fishery Products - Part 1: Frozen fish.<br>5.4 Determination of Thiobarbituric acid (TBA) |

### 1.5 Determination of ingredients in food using refractometric analysis [Flex A]

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|---|---|
| AOAC 921.08<br>2023-01 | Index of refraction of oils and fats |
| AOAC 932.14<br>2023-01 | Solids in syrups |

### 1.6 Determination of methyl esters of fatty acids in food using gas chromatography with conventional detectors (GC-FID) [Flex A]

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|---|---|
| ISO 12966-4<br>2015-06 | Animal and vegetable fats and oils - Gas chromatography of fatty acid methyl esters - Part 4: Determination by capillary gas chromatography |

### 1.7 Determination of ingredients, additives, pesticide residues, mycotoxins, contaminants in food and feed using high performance liquid chromatography with conventional detectors (UV, DAD, FLD, RI) [Flex B]

| | |
|---|---|
| AOAC 977.20<br>2023-01 | Separation of sugars in honey<br>(Modification: *extension for fructose, glucose and sucrose in biscuits*) |
| ISO 19343<br>2017-06 | Microbiology of the food chain - Detection and Quantification of Histamine in fish and fishery products - HPLC method |
| DIN EN 14123<br>2008-03 | Foodstuffs - Determination of aflatoxin $B_1$ and the sum of aflatoxin $B_1$, $B_2$, $G_1$ and $G_2$ in hazelnuts, peanuts, pistachios, figs and paprika powder - High performance liquid chromatographic method with post-column derivatisation and immunoaffinity column cleanup<br>(Restriction: *here only for nuts*) |
| CEN/TS 13130-13<br>2005-09 | Materials and articles in contact with foodstuffs - Plastics substances subject to limitation - Part 13: Determination of 2,2bis(4hydroxyphenyl) propane (Bisphenol A) in food simulants<br>(Modification: *here applied to edible oils*) |

| | |
|---|---|
| CEN/TS 13130-27<br>2005-02 | Materials and articles in contact with foodstuffs - Plastics substances subject to limitation - Part 27: Determination of 2,4,6-triamino-1,3,5-triazine in food simulants<br>(Restriction: *only in 3% acetic acid*) |
| Yu Y., Jin, Q., Wang X. G.<br>Chapter 5(10)<br>Pages 409-414<br>Journal of Chemical and Pharmaceutical Research, 2013 | Rapid assay of polycyclic aromatic hydrocarbon in edible oil by HPLC with FLD detection without cleanup |
| SOP-AT-02<br>2025-06 | Determination of Ochratoxin A in cereals by HPLC with FLD detection and immunoaffinity column clean-up |
| SOP-AT-04<br>2025-06 | Determination of additives (caffeine, benzoic acid, sorbic acid, acesulfame k, saccharin, ascorbic acid) in soft drinks and food by HPLC |
| SOP-AT-07<br>2025-06 | Detection and determination of artificial food dyes in beverages by using HPLC |
| SOP-AT-08<br>2025-06 | Detection and determination of para red and sudan dyes in chili powder products by HPLC |
| SOP-AT-09<br>2025-06 | Determination of tocopherols & tocotrienols in edible oil and fats by HPLC |
| SOP-AT-10<br>2025-06 | Determination of capsaicin in chili pepper products by HPLC |
| SOP-AT-11<br>2025-06 | Determination of zearalenone in cereal grains by HPLC |
| SOP-AT-12<br>2025-06 | Determination of deoxynivalenol (DON) in cereal by HPLC |
| SOP-AT-13<br>2025-06 | Determination of aflatoxin $M_1$ in powdered and liquid milk by HPLC |
| SOP-AT-16<br>2025-06 | Determination of common antioxidants (PG, TBHQ, BHA and BHT) in edible oil using HPLC |
| SOP-FCM-05<br>2025-06 | Determination of terephthalic acid in food simulants (3% acetic acid) using HPLC |

| | |
|---|---|
| SOP-TFA-17<br>2025-06 | Determination of sodium metabisulfite in food using HPLC |

**1.8 Determination of ingredients in food contact materials using high performance liquid chromatography with conventional detectors (UV, DAD, FLD, RI) [Flex B]**

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|---|---|
| SOP-FCM-07<br>2025-06 | Determination of isocyanates (2,4-toluene diisocyanate, 2,6-toluene diisocyanate, diphenylmethane-4,4'-diisocyanate, hexamethylene diisocyanate) in plastics using HPLC |

**1.9 Determination of pesticide residues in food using gas chromatography with mass selective detectors (MS/MS) [Flex B]**

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|---|---|
| EN 15662<br>2018-05 | Foods of plant origin - Multimethod for the determination of pesticide residues using GC- and LC-based analysis following acetonitrile extraction/partitioning and clean-up by dispersive SPE-Modular QuEChERS-method |

**1.10 Determination of contaminants in food contact materials using gas chromatography with mass selective detectors (MS/MS) [Flex B]**

| | |
|---|---|
| SOP-FCM-09<br>2025-06 | Determination of phthalates (DBP, DEHP, BBP, DINP, DIDP, DNOP) in food simulants by triple quadrupole GC-MS/MS |

**1.11 Determination of pesticide residues, mycotoxins and contaminants in food using high performance liquid chromatography with mass selective detectors (MS/MS, HR-MS) [Flex B]**

| | |
|---|---|
| EN 15662<br>2018-05 | Foods of plant origin - Multimethod for the determination of pesticide residues using GC- and LC-based analysis following acetonitrile extraction/partitioning and clean-up by dispersive SPE-Modular QuEChERS-method |
| EURL-SRM QuPPe-PO V12.3<br>2024-12 | Quick Method for the analysis of highly polar Pesticides in Food Involving Extraction with Acidified Methanol and LC-MS/MS Measurement Food of Plant Origin (QuPPe-PO-Method)<br>(Restriction: *here only methods M1.3 and M1.5*) |
| SOP-AT-14<br>2025-06 | Detection and quantification of histamine in fish and fishery products - LC-MS/MS method |
| SOP-TFA-18<br>2025-06 | Determination of mycotoxins (aflatoxins B1, B2, G1, G2, M1, OTA, DON, ZON, Fuminosin B1, B2, T2-toxin, patulin) by liquid chromatography tandem mass spectrometry (LC-MS/MS) in food |

## 1.12 Determination of elements in food, feed and food contact materials using atomic absorption spectrometry (GF-AAS, F-AAS) [Flex B]

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|---|---|
| AOAC 975.03<br>2023-01 | Metals in plants and pet foods - atomic absorption spectrophotometric method<br>(Modification: *here determination of iron, copper, zinc, magnesium, lead and cadmium in food and feed*) |
| AOAC 999.11<br>2023-01 | Lead, cadmium, copper, iron and zinc in foods - Atomic absorption spectrophotometry after dry ashing |
| SM 3111B<br>2017 | Metals by flame atomic absorption spectrometry. Part B: Direct Air-Acetylene Flame Method |
| ES 2060-01<br>2008-08 | Release of lead and cadmium from ware in contact with food and beverages. Part 1: Glass hollow ware |
| ES 2060-02<br>2007-06 | Release of lead and cadmium from ware in contact with food and beverages. Part 2: Ceramic ware |

## 1.13 Determination of elements using inductive coupled plasma-optical emission spectroscopy (ICP-OES), inductive coupled plasma-mass spectrometry (ICP-MS) and mercury analyzer [Flex A]

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|---|---|
| EN 12498<br>2018-11 | Paper and board - Paper and board intended to come into contact with foodstuffs - Determination of cadmium, chromium and lead in an aqueous extract<br>(Modification: *extension for aluminum*) |
| EN 15763<br>2009-12 | Foodstuffs - Determination of trace elements - Determination of arsenic, cadmium, mercury and lead in foodstuffs by inductively coupled plasma mass spectrometry (ICP-MS) after pressure digestion<br>(Modification: *here only arsenic, cadmium and lead*) |
| EN 15765<br>2009-12 | Foodstuffs - Determination of trace elements - Determination of tin by inductively coupled plasma mass spectrometry (ICP-MS) after pressure digestion |
| EN 16943<br>2017-05 | Foodstuffs - Determination of calcium, copper, iron, magnesium, manganese, phosphorus, potassium, sodium, sulfur and zinc by ICP-OES<br>(Modification: *here only copper, iron, zinc; extension for arsenic, cadmium, lead and tin*) |

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|---|---|
| EU Regulation 10/2011<br>2011-01 | Commission Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food |
| EPA 7473<br>2007-02 | Mercury in solids and solutions by thermal decomposition, amalgamation and atomic absorption spectrophotometry<br>(Modification: *analysis carried out using a mercury analyzer, here for food*) |
| SOP-TE-07<br>2025-06 | Determination of trace elements (Sn, Pb, Cd, As) in food samples by ICP-MS after microwave digestion |

**1.14 Determination of the nitrogen content in Food using Combustion Analysis FLASH Smart N/ Protein**

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|---|---|
| SOP-CH-17<br>2025-06 | Determination of protein & nitrogen in food by flash smart elemental analyzer |

**2 Cultural microbiological analysis of bacteria, yeasts, molds in food, feed, environmental samples, equipment and utensils in the sector of food [Flex B]**

| | |
|---|---|
| ISO 4832<br>2006-02 | Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of coliforms - Colony-count technique |
| ISO 4833-1 AMD 1<br>2022-01 | Microbiology of the food chain - Horizontal method for the enumeration of microorganisms - Part 1: Colony count at 30 degrees C by the pour plate technique |
| ISO 6579-1 AMD 1<br>2020-03 | Microbiology of food chain - Horizontal method for the detection, enumeration and serotyping of Salmonella spp.<br>(Modification: *here only detection*) |
| ISO 6888-1 AMD 1<br>2023-09 | Microbiology of food chain - Horizontal method for the enumeration of coagulase-positive staphylococci (Staphylococcus aureus and other species) - Part 1: Method using Baird-Parker agar medium |
| ISO 7932-Amd 1<br>2020-08 | Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of presumptive Bacillus cereus - Colony-count technique at 30 °C |
| ISO 10272-1<br>2023-01 | Microbiology of the food chain - Horizontal method for the detection and enumeration of Campylobacter spp. - Part 1: Detection method |
| ISO 11290-1<br>2017-05 | Microbiology of food chain - Horizontal method for detection and enumeration of Listeria monocytogenes and of Listeria spp. -<br>Part 1: Detection method<br>(Modification: *Confirmation by Microbact 12L*) |

| | |
|---|---|
| ISO 15213 -2<br>2023 | Microbiology of the food chain - Horizontal method for the detection and enumeration of *Clostridium* spp. – Part 2: Enumeration of *Clostridium perfringens* by colony count technique |
| ISO 16649-2<br>2001-04 | Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of β-glucuronidase-positive Escherichia coli - Part 2: Colony-count technique at 44 °C using 5-bromo-4-chloro-3-indolyl β-D-glucoronide |
| ISO 18593<br>2018-06 | Microbiology of the food chain - Horizontal methods for surface sampling |
| ISO 21527-1<br>2008-07 | Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of yeasts and moulds - Part 1: Colony count technique in products with water activity greater than 0,95<br>(Modification: *here only determination of yeasts; 3 days, 30 °C*) |
| ISO 21527-2<br>2008-07 | Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of yeasts and moulds - Part 2: Colony count technique in products with water activity less than or equal to 0,95 |
| ISO 21528-2<br>2017-06 | Microbiology of food and animal feeding stuffs - Horizontal methods for the detection and enumeration of Enterobacteriaceae -<br>Part 2: Colony-count method |
| ISO 21567<br>2004-11 | Microbiology of food and animal feeding stuffs - Horizontal method for the detection of Shigella spp. In Food and animal feeding stuffs |
| ISO 21872-1<br>2023-02 | Microbiology of the food chain - Horizontal method for the determination of Vibrio spp. - Part 1: Detection of potentially enteropathogenic Vibrio parahaemolyticus, Vibrio cholerae and Vibrio vulnificus |
| ISO/TS 21872-2<br>2007-04 | Microbiology of food and animal feeding stuffs - Horizontal method for the detection of potentially enteropathogenic Vibrio spp. -<br>Part 2: Detection of species other than Vibrio parahaemolyticus and Vibrio cholera<br>(Modification: *detection of Vibrio alginolyticus*) |
| ISO 22964<br>2017-04 | Microbiology of the food chain - Horizontal method for the detection of Cronobacter spp. |
| NMKL No. 68<br>2011 | Enterococcus. Determination in foods and feeds. |

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|---|---|
| NMKL No. 125<br>2005 | Thermotolerant coliform bacteria and Escherichia coli. Enumeration in food and feed. |
| 3M™ Petrifilm™<br>Rapid Yeast and<br>Mold Count Plate<br>(6475/6477)<br>2017 | Enumeration of Mould by Petrifilm technique |

**3 Determination of bacteria and virus in food and feed using Real-Time-PCR [Flex B]**

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|---|---|
| ISO/TS 15216-2<br>2019-07 | Microbiology of the food chain - Horizontal method for determination of hepatitis A virus and norovirus in food using real-time RT-PCR, Part 2: Method for detection |
| Congen<br>Art. No. F5166<br>2019-07 | SureFast® Salmonella species/enteritidis/typhimurium 4plex |
| Hygiena<br>foodproof$^{R}$ Hepatitis A<br>Detection Kit<br>R 302 37<br>2017-09 | Test kit for the qualitative detection of hepatitis A virus RNA in food |
| Hygiena<br>foodproof$^{R}$ *E. coli 0157*<br>Detection Kit<br>R 302 10<br>2017-04 | Test kit for the qualitative detection of *Escherichia coli* of serogroup 0157 in food |
| BIO-RAD<br>– IQ –Check-Listeria<br>monocytogenes II Kit<br>Ref no. 3578124<br>2021-03 | Test kit for the qualitative detection of Listeria monocytogenes in food |

**Abbreviations used:**

| | |
|---|---|
| AOAC | Association of Official Analytical Chemists |
| AMD | Amendment |
| CEN | European committee for standardization |
| DIN | German Institute of Standardization |
| EN | European Standards |
| EPA | United States Environmental Protection Agency (EPA) |
| ES | Egyptian Standard |
| EURL-SRM | EU Reference Laboratory for Single Residue Methods |
| IEC | International Electrotechnical Commission |
| ISO | International Organization for Standardization |
| NMKL | Nordic Committee on Food Analysis |
| SM | Standard Methods For the Examination of Water and Wastewater™ |
| SOP-AT-xy | In-house method of the RIIL laboratories |